Name :- Mihir Mungara

Airflow : Assignment-2

Definition :- Create DAG which will take 7 different types input from user and display them through a task or create some usecase using few inputs.

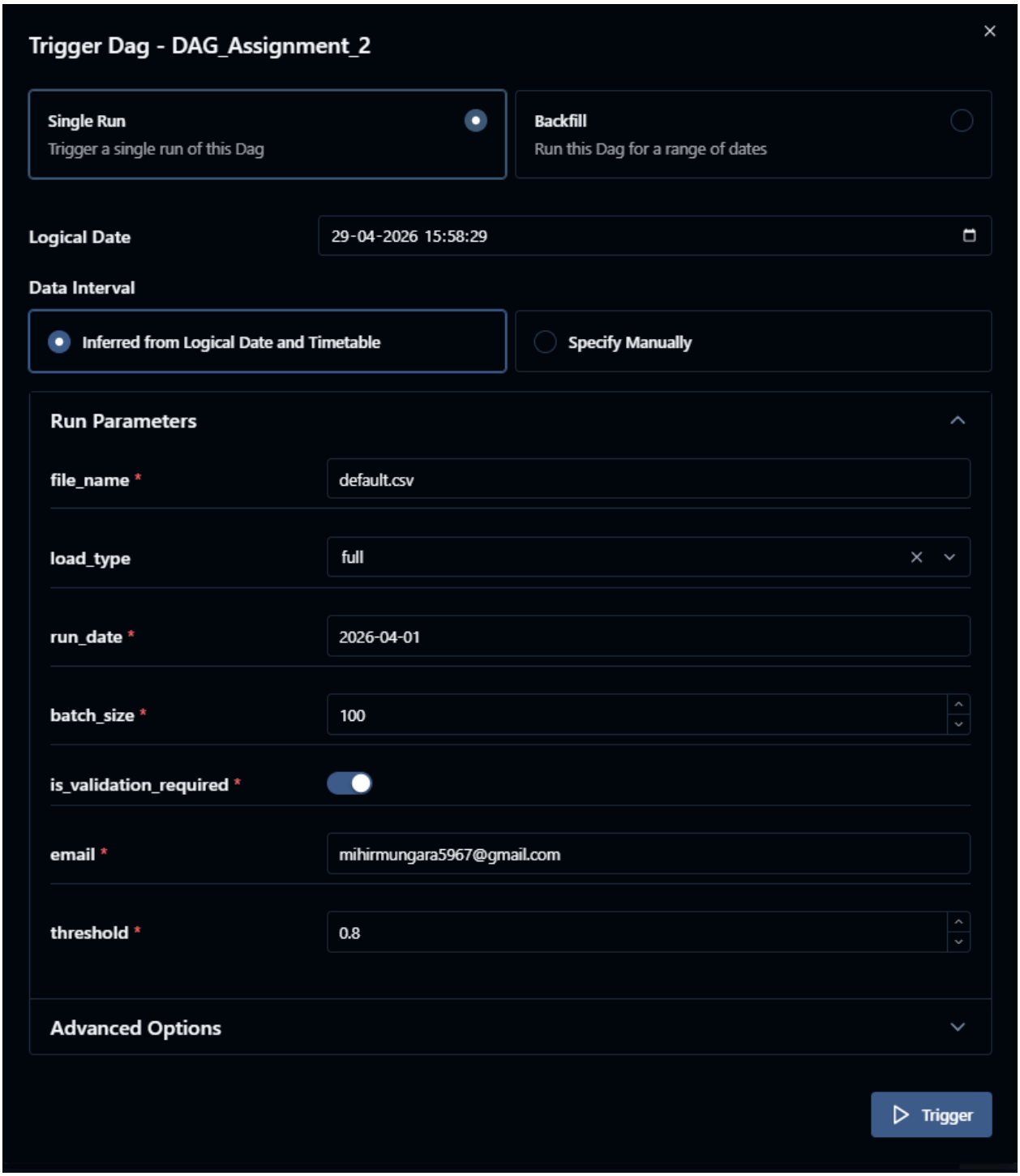
Dag File :-

```
from airflow import DAG
from airflow.models.param import Param
from airflow.decorators import dag, task
from datetime import datetime, timedelta, date

default_args = {
    'owner': 'mihir',
    'retries': 5,
    'retry_delay': timedelta(minutes=2)
}

@dag(
    dag_id='DAG_Assignment_2',
    default_args=default_args,
    description='Params example DAG',
    start_date=datetime(2026, 4, 25),
    schedule='@daily',
    catchup=False,
    params={
        'file_name': Param('default.csv', type='string'),
        'load_type': Param('full', enum=['full', 'incremental']),
        'run_date': Param("2026-04-01", type='string'),
        'batch_size': Param(100, type='integer', minimum=1),
        'is_validation_required': Param(True, type='boolean'),
        'email': Param('mihirmungara5967@gmail.com', type='string'),
        'threshold': Param(0.8, type='number')
    }
)
def assignment_dag():
    # Task 1: Print Params
    @task
    def print_params(params=None):
        print("File Name:", params['file_name'])
        print("Load Type:", params['load_type'])
```

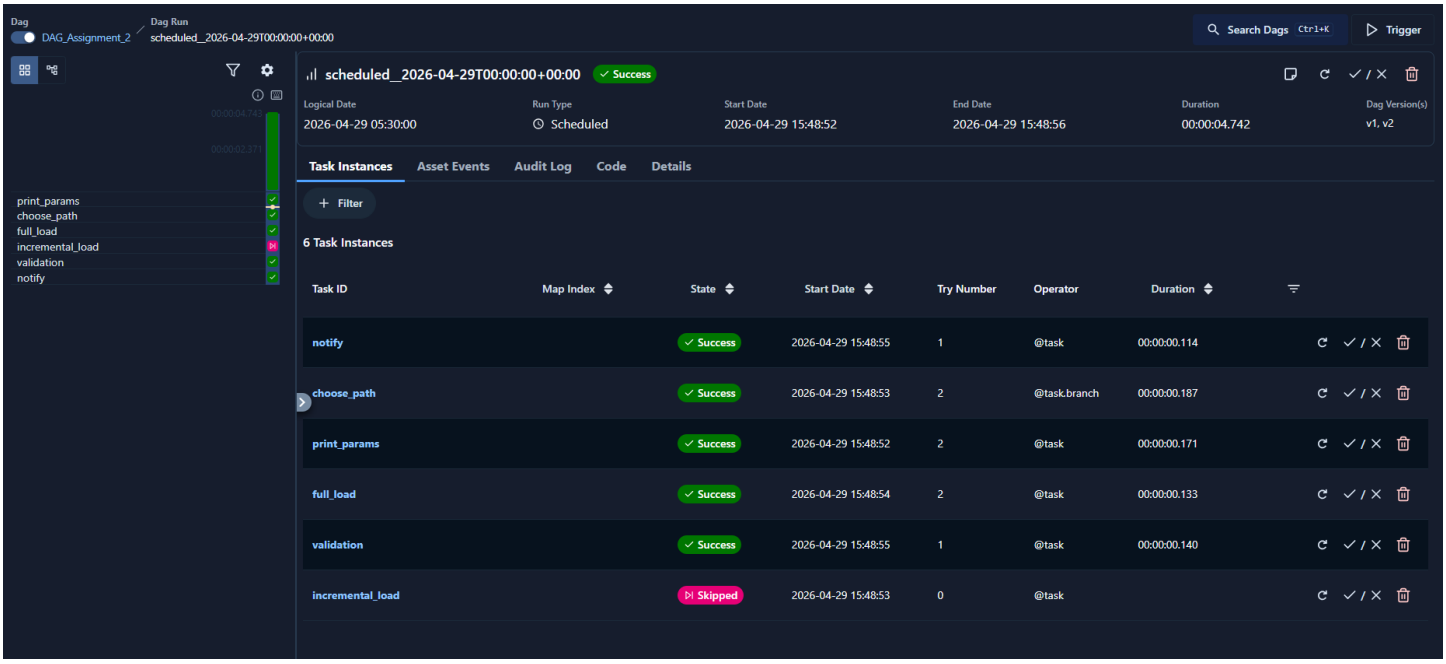
```python
    print("Run Date:", params['run_date'])
    print("Batch Size:", params['batch_size'])
    print("Validation Required:", params['is_validation_required'])
    print("Email:", params['email'])
    print("Threshold:", params['threshold'])

# Task 2: Branch Logic
@task.branch
def choose_path(params=None):
    if params['load_type'] == 'full':
        return 'full_load'
    else:
        return 'incremental_load'

# Task 3: Full Load
@task
def full_load():
    print("Running FULL load")

# Task 4: Incremental Load
@task
def incremental_load():
    print("Running INCREMENTAL load")

# Task 5: Validation
@task
def validation(params=None):
    if params['is_validation_required']:
        print(f"Running validation with threshold {params['threshold']}")
    else:
        print("Validation skipped")

# Task 6: Notification
@task
def notify(params=None):
    print(f"Sending notification to {params['email']}")

# Flow
p = print_params()
branch = choose_path()
f = full_load()
i = incremental_load()

v = validation.override(trigger_rule="one_success")()
n = notify.override(trigger_rule="one_success")()

p >> branch
branch >> f >> v >> n
```

```
    branch >> i >> v >> n

dag = assignment_dag()
```

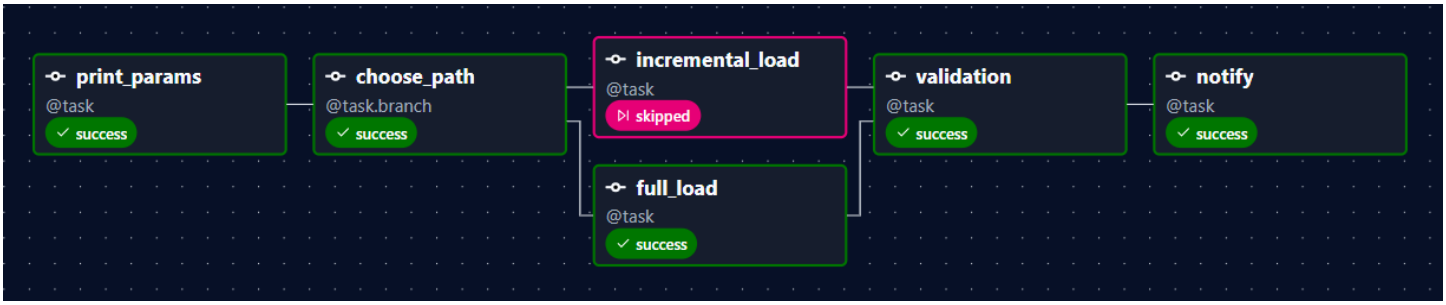
Input :-



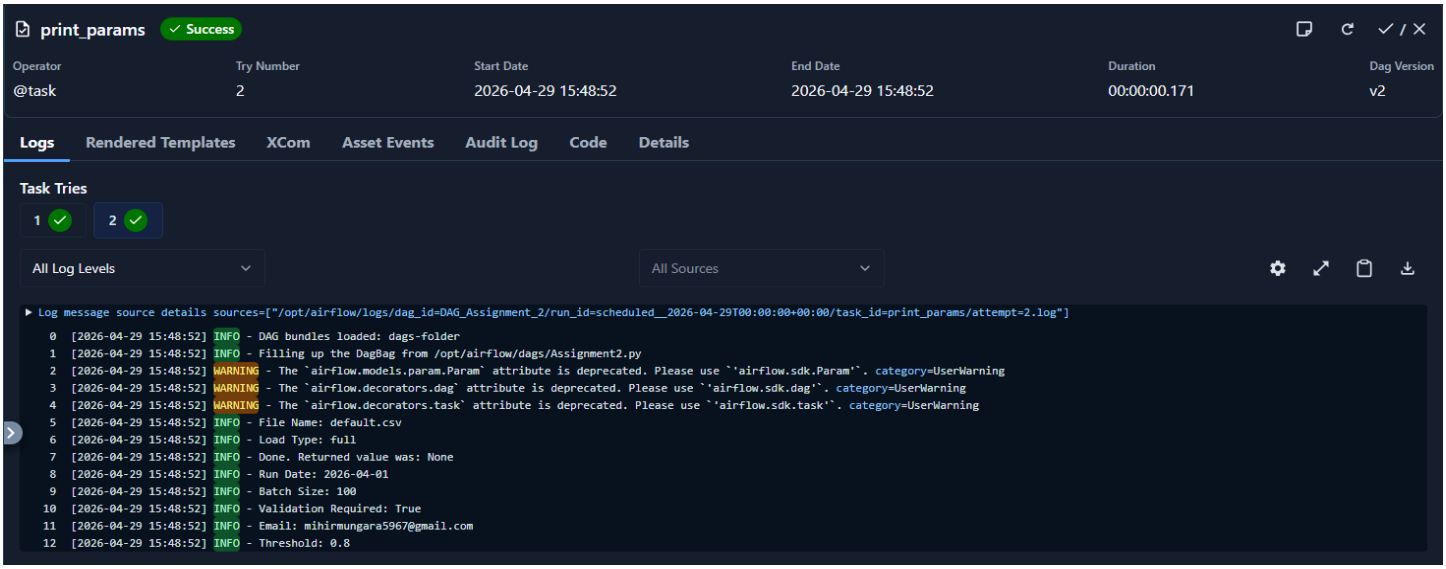
Results :-



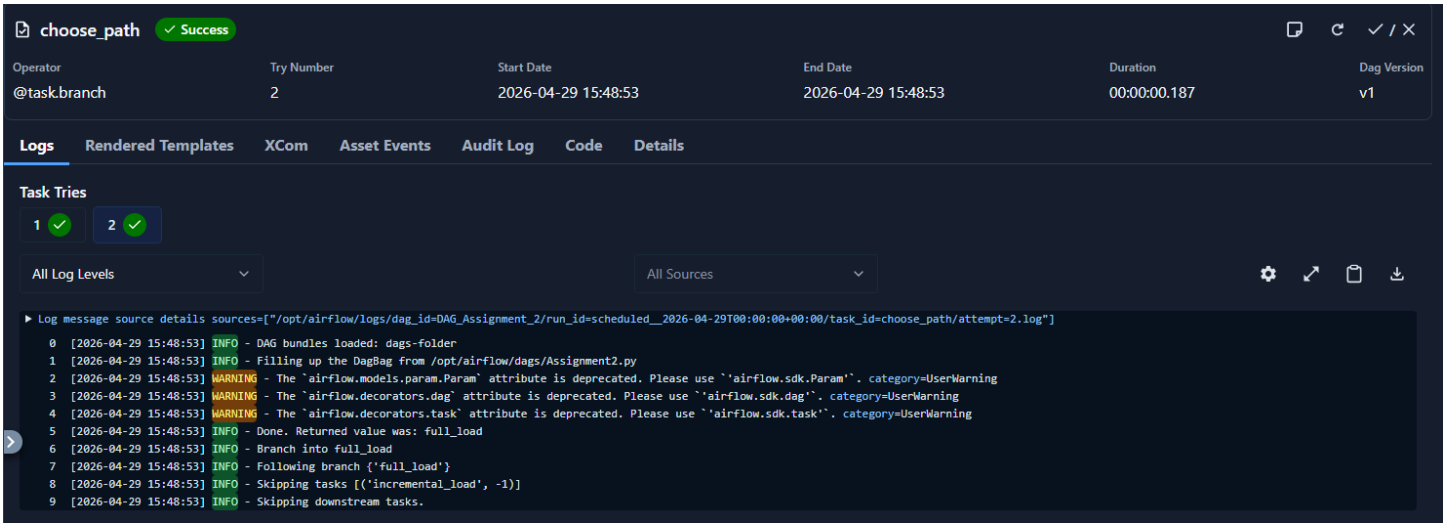
Graph View :-



Print_Params Logs :-

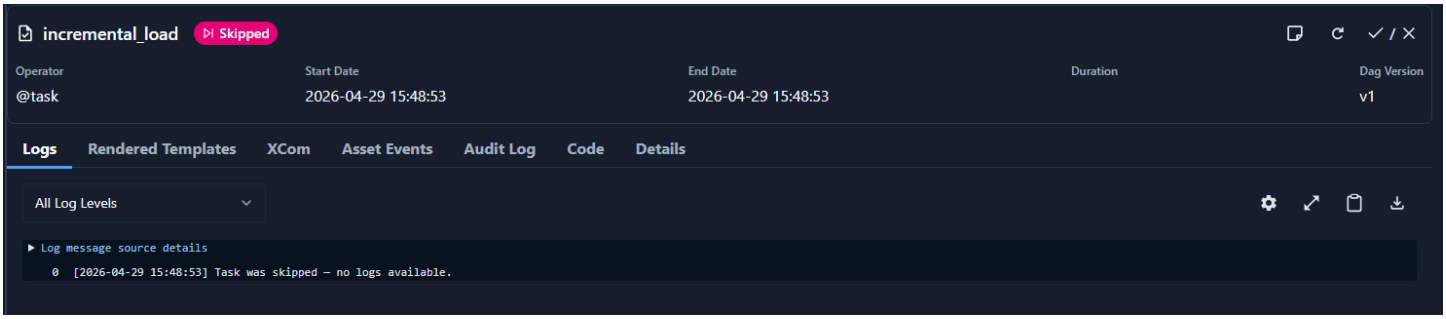


Choosing Branch Logs :-



Full Load Logs :-

Incremental Load Logs :-



Validation Logs :-

validation ✓ Success

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-04-29 15:48:55	2026-04-29 15:48:55	00:00:00.140	v1

Logs | Rendered Templates | XCom | Asset Events | Audit Log | Code | Details

All Log Levels | All Sources

```
▶ Log message source details sources=["/opt/airflow/logs/dag_id=DAG_Assignment_2/run_id=scheduled__2026-04-29T00:00:00+00:00/task_id=validation/attempt=1.log"]
  0  [2026-04-29 15:48:55] INFO - DAG bundles loaded: dags-folder
  1  [2026-04-29 15:48:55] INFO - Filling up the DagBag from /opt/airflow/dags/Assignment2.py
  2  [2026-04-29 15:48:55] WARNING - The `airflow.models.param.Param` attribute is deprecated. Please use `'airflow.sdk.Param'`. category=UserWarning
  3  [2026-04-29 15:48:55] WARNING - The `airflow.decorators.dag` attribute is deprecated. Please use `'airflow.sdk.dag'`. category=UserWarning
  4  [2026-04-29 15:48:55] WARNING - The `airflow.decorators.task` attribute is deprecated. Please use `'airflow.sdk.task'`. category=UserWarning
  5  [2026-04-29 15:48:55] INFO - Done. Returned value was: None
  6  [2026-04-29 15:48:55] INFO - Running validation with threshold 0.8
```

Notify Logs :-

notify ✓ Success

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-04-29 15:48:55	2026-04-29 15:48:56	00:00:00.114	v1

Logs | Rendered Templates | XCom | Asset Events | Audit Log | Code | Details

All Log Levels | All Sources

```
▶ Log message source details sources=["/opt/airflow/logs/dag_id=DAG_Assignment_2/run_id=scheduled__2026-04-29T00:00:00+00:00/task_id=notify/attempt=1.log"]
  0  [2026-04-29 15:48:55] INFO - DAG bundles loaded: dags-folder
  1  [2026-04-29 15:48:55] INFO - Filling up the DagBag from /opt/airflow/dags/Assignment2.py
  2  [2026-04-29 15:48:56] WARNING - The `airflow.models.param.Param` attribute is deprecated. Please use `'airflow.sdk.Param'`. category=UserWarning
  3  [2026-04-29 15:48:56] WARNING - The `airflow.decorators.dag` attribute is deprecated. Please use `'airflow.sdk.dag'`. category=UserWarning
  4  [2026-04-29 15:48:56] WARNING - The `airflow.decorators.task` attribute is deprecated. Please use `'airflow.sdk.task'`. category=UserWarning
  5  [2026-04-29 15:48:56] INFO - Sending notification to mihirmungara5967@gmail.com
  6  [2026-04-29 15:48:56] INFO - Done. Returned value was: None
```